

CREDIT RISK SCORECARD MODEL

Indian Lending Portfolio Analysis

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Purpose: Portfolio Project | Interview Preparation

Built a complete Credit Risk Scorecard with 80 realistic Indian borrower profiles, scoring model (0-100), risk-based pricing, scenario testing, and interactive dashboard.

1. EXECUTIVE SUMMARY

As part of my MBA Finance portfolio, I developed a complete **Credit Risk Scorecard Model** for evaluating Indian borrowers. The model processes 80 realistic applicant profiles and provides a structured way to assess risk, make approval decisions, and suggest appropriate interest rates using risk-based pricing. **Key Deliverables:**

- 5-sheet Excel model with Raw Data, Calculations, Credit Scorecard, Decision Engine, and Scenario Analysis
 - Scoring system (0-100) based on Income, DTI Ratio, Credit History, and Employment Type
 - Risk classification: Low Risk (Approve), Medium Risk (Conditional), High Risk (Reject)
 - Risk-based interest rates ranging from 9.75% to 15.5%
 - Scenario testing for income drop and EMI increase
 - Live interactive dashboard (Power BI + Streamlit)
- Main Finding:** In the Indian lending context, **DTI Ratio above 35%** and **Poor Credit History** are the strongest predictors of high risk and rejection. The model shows that even a 20% drop in income can push many Medium Risk borrowers into the High Risk category.

2. MODEL OVERVIEW & DATA

Objective: Create a practical, interview-ready Credit Risk Scorecard that banks and NBFCs in India can use to evaluate personal loan / consumer loan applicants. **Data:** 80 realistic borrower records generated with Indian context (Metro, Tier-1, Tier-2, Tier-3 regions). Variables include Monthly Income (■15k–■1.8L), Loan Amount, Existing EMI, Credit History (Good/Average/Poor), Employment Type (Salaried/Self-Employed), Loan Tenure, Purpose, and Region. **Key Calculations:**

- **DTI Ratio** = $(\text{Existing EMI} \div \text{Monthly Income}) \times 100$

- **Loan Burden** = $\text{Loan Amount} \div (\text{Monthly Income} \times 36)$

- **Income Multiple** = $\text{Loan Amount} \div \text{Monthly Income}$ These indicators are commonly used by Indian lenders and are easy to explain in interviews.

3. CREDIT SCORECARD LOGIC

I designed a transparent 0–100 point scoring system. Higher score = Lower Risk. **Scoring Breakdown:**

- **Monthly Income (Max 35 pts):** Higher income → higher score (reflects repayment capacity)
- **DTI Ratio (Max 30 pts):** Lower DTI → much higher score (most important risk driver in India)
- **Credit History (Max 25 pts):** Good = 25, Average = 12, Poor = 3 (strongest predictor)
- **Employment Type (Max 10 pts):** Salaried = 10, Self-Employed = 5 (stable income preferred) **Final**

Classification:

- **75–100 (Low Risk) → APPROVE** at 9.75% – 11.5%
- **55–74 (Medium Risk) → CONDITIONAL APPROVAL** at 12% – 15.5% (with collateral/guarantor/monitoring)
- **<55 (High Risk) → REJECT Risk-Based Pricing Logic:** Higher risk borrowers pay higher interest rates. This is standard practice in Indian NBFCs and banks to compensate for higher expected losses. It also encourages better borrower behavior.

4. POWER BI / LIVE DASHBOARD

I built an interactive dashboard with the following features: **Key Performance Indicators (KPIs):**

- Total Applicants: 80
- Approval Rate: ~64%
- High Risk Percentage: ~21%
- Average Credit Score: 68.4 (Medium Risk zone)

Visuals Included:

- Risk Distribution (Pie chart – Low / Medium / High)
 - Income vs Credit Score scatter plot (shows clear relationship)
 - Scenario Impact table (What-if analysis)
 - Filters: Income Range, Risk Category, Employment Type
- The dashboard is built in both **Power BI** (industry standard) and **Streamlit** (for live web deployment). It allows quick filtering and stress testing — very useful for credit officers and risk managers.

5. KEY INSIGHTS

Here are the most important learnings from building and testing this model: **1. DTI Ratio above 35% is the biggest red flag**

Applicants with high existing debt relative to income had significantly higher rejection rates. In Indian lending, existing EMI burden often matters more than current income level. **2. Credit History is the strongest single predictor**

I gave it the highest weight (25 points) because in India, past repayment behavior (CIBIL score, previous defaults) is the #1 indicator of future default probability. **3. Salaried vs Self-Employed makes a real difference**

72% of Low Risk cases were Salaried applicants. Self-employed borrowers need stronger credit history or collateral to get approved at good rates because of income volatility. **4. Small changes can flip risk categories**

A 20% income drop or 30% increase in EMI pushed many Medium Risk borrowers into High Risk. This is exactly why banks conduct stress testing during economic slowdowns or rate hike cycles. **5. Risk-based pricing is fair and practical**

Charging 13–15.5% to Medium Risk borrowers compensates the lender for higher expected loss while still giving the borrower access to credit. This is standard RBI-compliant practice. **6. Metro + Salaried + Good Credit = Almost automatic approval**

This combination consistently scored above 80. The model confirms what credit officers observe daily in real lending. **7. The model is transparent and explainable**

Every score and decision can be explained with clear logic. This is very important during interviews and also for regulatory compliance (fair lending practices).

6. FINAL CONCLUSION

This Credit Risk Scorecard Model project helped me understand how banks and NBFCs in India actually assess consumer credit risk. Building the model from scratch — from data generation to scoring logic to scenario testing — gave me practical, interview-ready skills in financial modeling, risk analysis, and data visualization. **What I Learned:**

- How to structure a real-world credit decisioning system
 - Importance of DTI and Credit History in Indian lending
 - How to use scenario analysis for stress testing
 - How to communicate complex risk concepts simply (very useful in interviews)
- Skills Demonstrated:**

Advanced Excel, Power BI, Risk Modeling, Scenario Analysis, Data Visualization, Investment Decision Making

Future Scope:

The model can be expanded with machine learning (Logistic Regression / Random Forest) for probability of default prediction, or integrated with actual bureau data (CIBIL) for production use. This project, along with my IRCTC Financial Model and certifications, has significantly strengthened my profile for roles in Credit Risk, Corporate Banking, Investment Banking, and Equity Research. I am happy to walk through the full model (Excel + Dashboard) in any interview or discussion.

— End of Report —

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